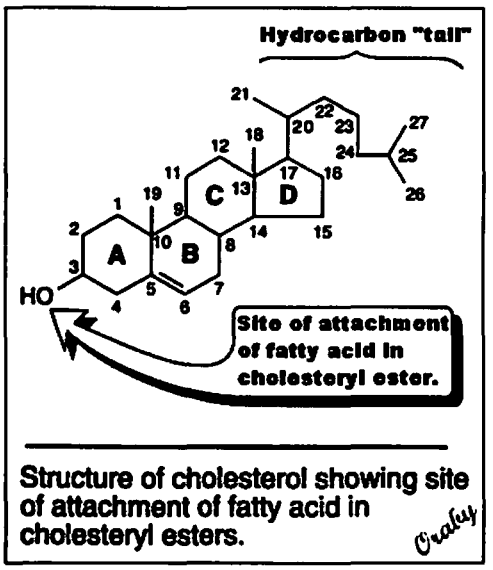
Metabolism of Cholesterol

I. STRUCTURE:

A. Cholesterol is an animal sterol.

B. It is a solid alcohol having –OH group at C_3.



Structure of cholesterol showing site of attachment of fatty acid in cholesteryl esters.

II. SOURCES OF CHOLESTEROL:

A. **Endogenous**: Cholesterol is formed in the body almost in all nucleated cells from Acetyl-CoA (about 700 mg/day).

B. **Exogenous**: Cholesterol occurs only in food of animal origin such as egg yolk, meat, liver and brain. Diet supplies about 400 mg/day.

III. SYNTHESIS OF CHOLESTEROL:

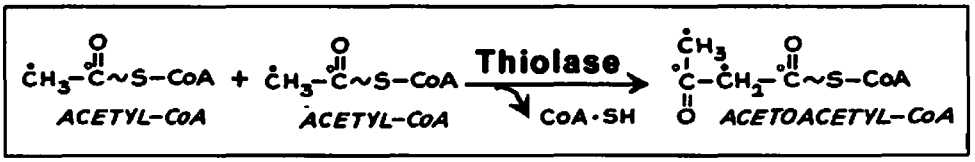
A. **Location**:

1. **Intracellular location**: Cytosol.
2. **Organ location**:
   a) **Liver** is the major site of cholesterol synthesis.
   b) Other tissues e.g. intestine, adrenal cortex, gonads and skin.

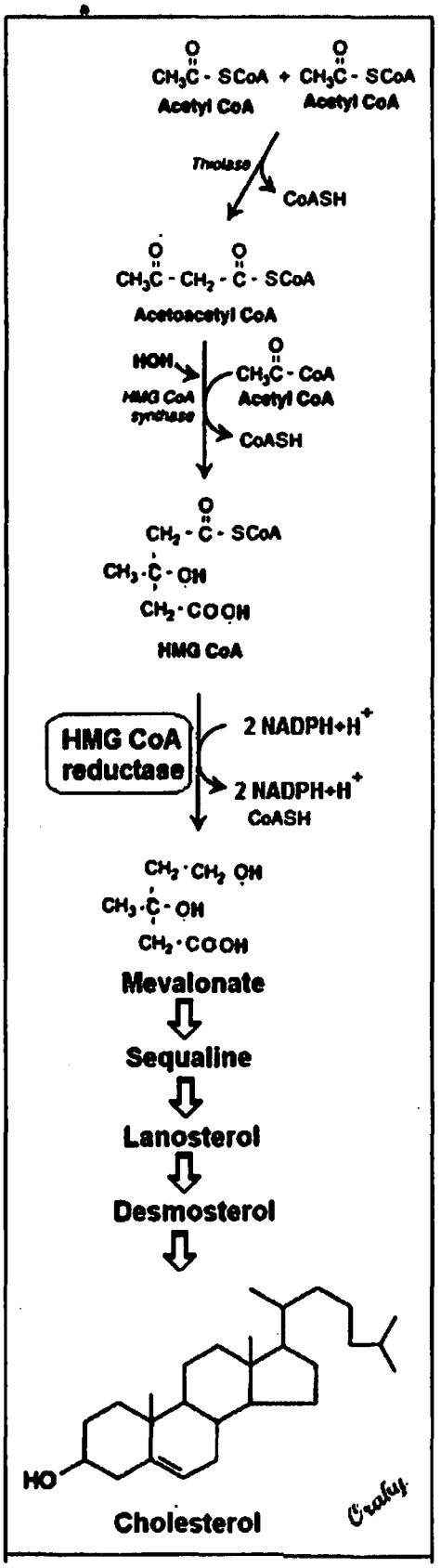
B. **Precursor**: Acetyl-CoA.

C. **Steps**:

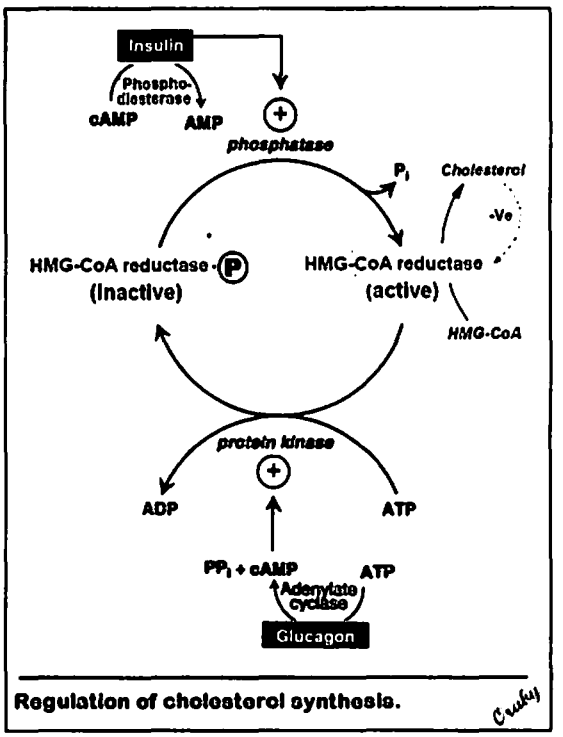
1. **Formation of acetoacetyl CoA**: by condensation of two molecules of acetyl CoA:



2. Conversion of acetoacetyl CoA to cholesterol.

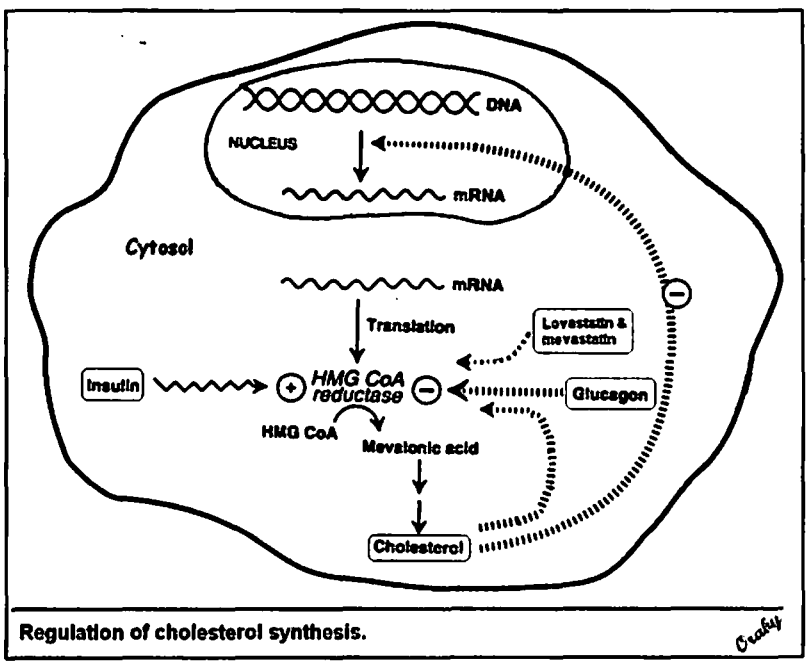


D. <u>Regulation of cholesterol synthesis</u>: ***HMG CoA reductase*** is the key enzyme for cholesterol synthesis. It is present in two forms: active dephosphorylated and inactive phosphorylated. It is Regulated through:



Regulation of cholesterol synthesis.

1. **Feed back Inhibition:** Cholesterol *(the end product of the pathway)* acts as feed back inhibitor of HMG CoA reductase enzyme. Thus, it decreases more cholesterol synthesis.
2. **Feed back regulation:** Cholesterol *(either synthesized by the cell or reaching it from diet)* inhibits HMG CoA reductase **<u>gene</u>**. This decreases transcription and synthesis of HMG CoA reductase.
3. **Hormonal regulation:**
   a) **Glucagon:** Inhibits HMG CoA reductase (through stimulation of cAMP and protein kinase).
   b) **Insulin:** Stimulates HMG CoA reductase (through stimulation of phosphodiesterase and phosphatase enzymes).
4. **Inhibition by drugs:** Lovastatin and mevastatin are drugs, which inhibit HMG CoA reductase by reversible competitive inhibition. They are used to decrease plasma cholesterol levels in patients with hyper-cholesterolemia.



Regulation of cholesterol synthesis.

IV. *FUNCTIONS OF CHOLESTEROL:*

A. Cholesterol enters in the structure of **every body cell** (e.g. cell membrane).

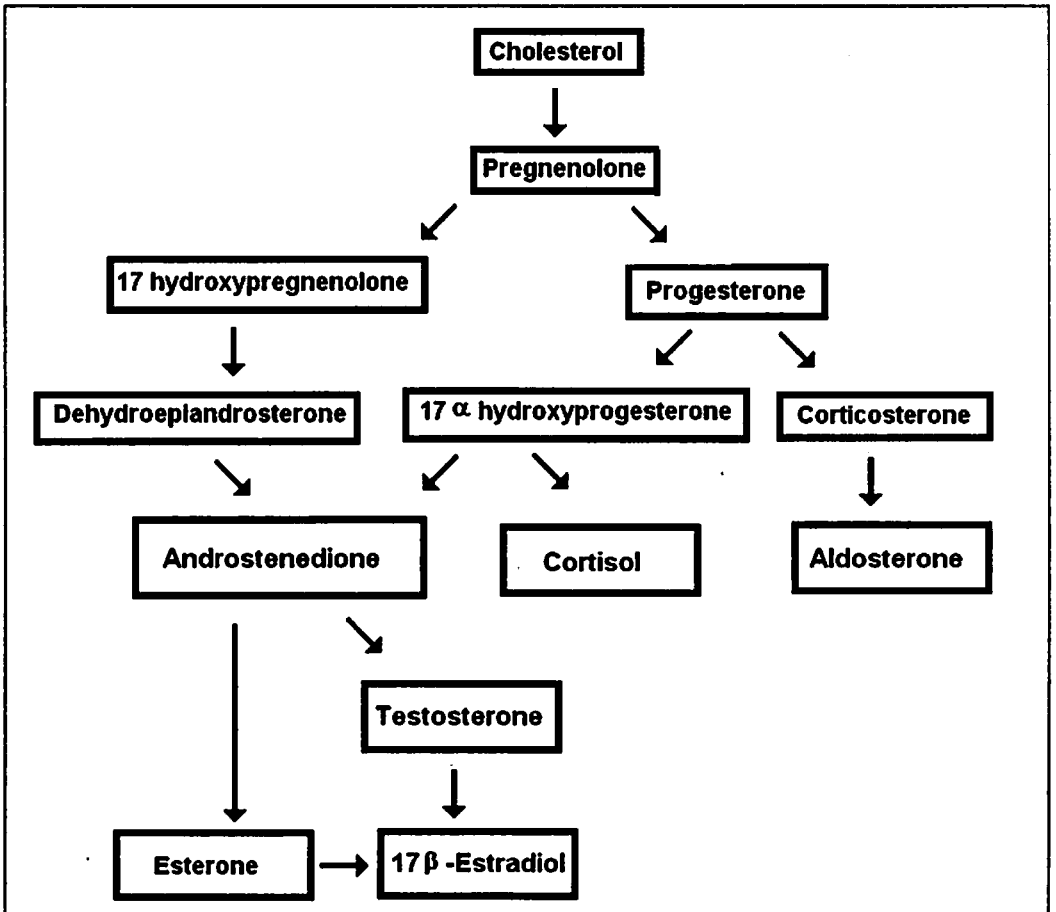
B. **Cholesterol is the precursor of:**

1. **Vitamin D_3:** (subcutaneous fat) ➔ 7-dehydrocholesterol ➔ Vitamin D_3.
2. **Steroid hormones:**
   a) **Estrogens and progesterone** (ovaries).
   b) **Testosterone and androgens** (testes).
   c) **Glucocorticoids and mineralocorticoids** (Adrenal cortex).
3. **bile acids: (Liver)**

1. **Synthesis of vitamin D_3:** See part I, vitamins.

Cholesterol ➔ 7-Dehydrocholesterol ➔ Vitamin D3 ***(SKIN)*** ➔ 25 hydroxy D3 ***(LIVER)*** ➔ 1,25 dihydroxy D3 ***(KIDNEY).***

2. **Synthesis of steroid hormones** (androgens, estrogens, progesterone and corticoids):

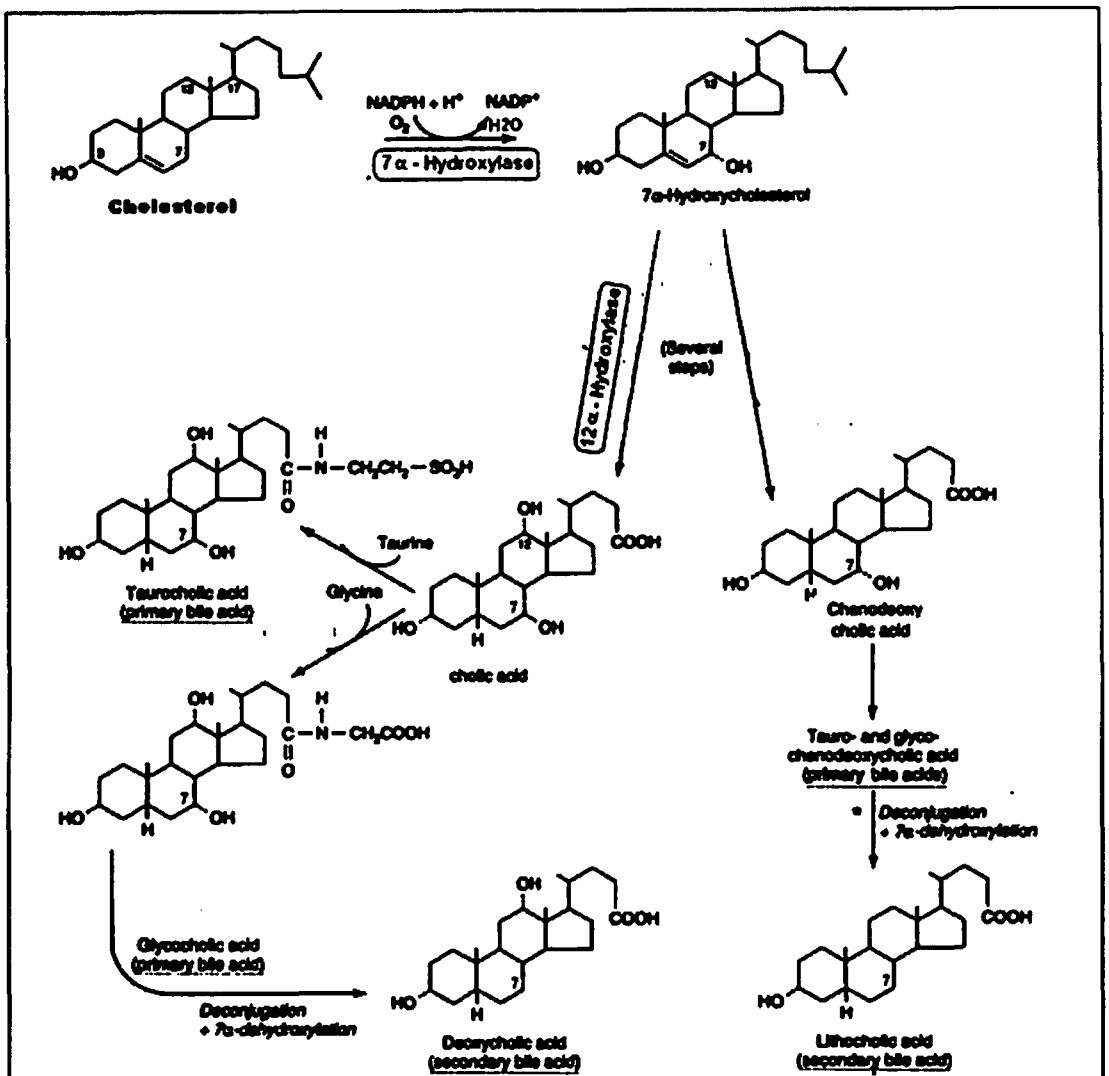


The biosynthesis of steroid hormones from cholesterol. Oraby

3. Synthesis of bile acids and salts:

a) Primary bile acids are cholic and chenodeoxy cholic acid.

b) Secondary bile acids are deoxycholic acid and lithocholic acid.



Biosynthesis and degradation of bile acids.
(Note that primary bile acids have an -OH at C7, while secondary bile acids have no -OH at C7).

V. *EXCRETION OF CHOLESTEROL:* About **one gram** of cholesterol is excreted **daily.** It is secreted as cholesterol, bile acids and coprostanol:

A. ½ Gram cholesterol is excreted as such with bile, which transports it to the intestine for elimination.

B. ½ Gram cholesterol is converted to bile acids, which is excreted in the feces.

C. Some cholesterol is synthesized by intestinal cells and modified by bacteria before excretion. Bacterial enzymes reduce cholesterol into **coprostanol**, which is excreted into feces.

VI. *PLASMA CHOLESTEROL :*

A. Cholesterol present in plasma is either free or esterified (cholesteryl ester).
   1. Total plasma cholesterol: 140 –220 mg/dl.
   2. Free plasma cholesterol: 26 – 126 mg/dl.

B. **Hypercholesterolemia:**
   1. **Definition:** It is increased plasma cholesterol concentration above 220 mg/dl.
   2. **Causes:**
      a) **Diet** rich in carbohydrate, cholesterol and saturated fatty acids.
      b) **Hypothyroidism** as thyroxin stimulates conversion of cholesterol to bile acids.
      c) **Diabetes mellitus.**
      d) **Kidney affection** (nephrotic syndrome) unknown mechanism.
      e) **Obesity.**
      f) **Obstructive jaundice** due to decreased excretion of cholesterol and bile acids.
      g) **Familial hypercholesterolemia.**

C. **Hypocholesterolemia:**
   1. **Definition:** It is decreased plasma cholesterol concentration below 140 mg/dl
   2. **Causes :**
      a) **Prolonged fasting** which causes decreased secretion of insulin (decreased activation of HMG-CoA reductase).
      b) **Diet** rich in unsaturated fatty acids and poor in saturated fatty acids, carbohydrate and cholesterol.
      c) **Liver diseases,** as liver is the site where most plasma cholesterol is synthesized.
      d) **Hyperthyroidism.**
      e) **Chronic infection** as tuberculosis.

VI. *TRANSPORT OF CHOLESTEROL:*

A. Cholesterol is hydrophobic. It is transported in plasma in the more soluble lipoprotein forms: LDL, VLDL and HDL (see plasma lipoproteins).

B. Free cholesterol is removed from tissues by HDL and transported to be excreted by the liver.

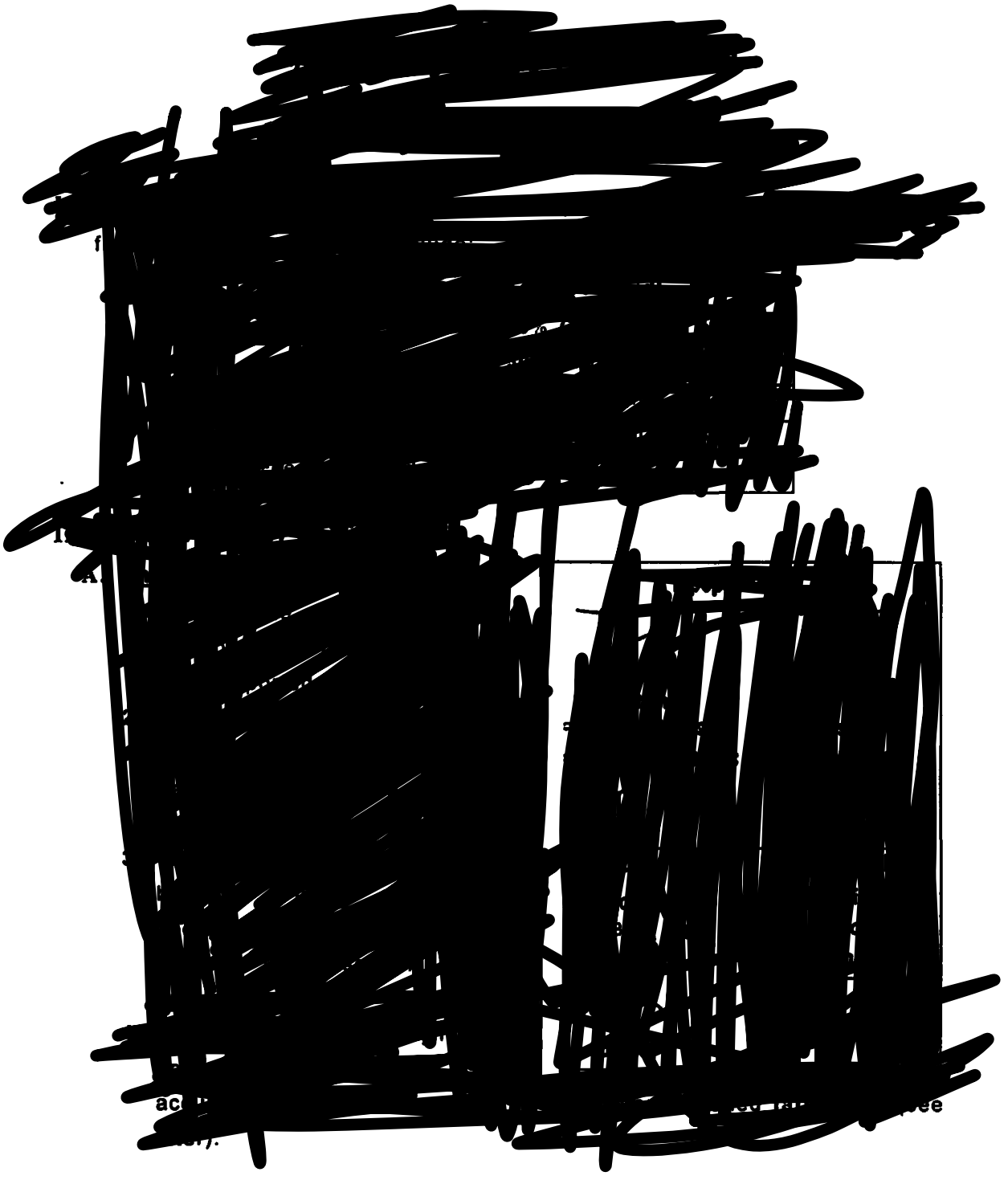
C. Cholesterol ester is the storage form of cholesterol: It is formed in both tissues and plasma.

1. In tissues (liver), cholesterol is esterified by **ACAT enzyme** (acyl CoA cholesterol acyl transferase):

Cholesterol + Acyl CoA ➔ Cholesteryl ester + CoASH

2. In plasma, cholesterol is esterified by **LCAT enzyme** (lecithin cholesterol acyl transferase). LCAT is associated with HDL.

Cholesterol + lecithin ➔ Cholesteryl ester + lysolecithin



Fatty liver

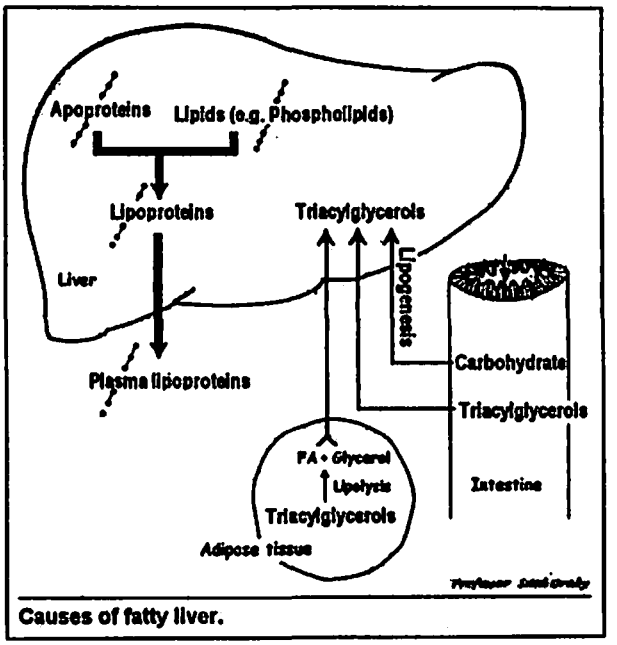
I. ***Definition of fatty liver:*** This is an accumulation of abnormal amount of fat in the liver for a long time with subsequent compression of liver cells. This results in liver fibrosis and impairment of liver function.

II. ***Causes:***

- Overmobilization of fat from extrahepatic tissues to the liver.
- High carbohydrate diet.
- Undermobilization of fat from the liver to the plasma.

A. **Causes of overmobilization of fat from extrahepatic tissues to the liver:**

1. After high fat diet.
2. Excessive lipolysis due to low carbohydrate diet, starvation and diabetes mellitus.



Causes of fatty liver.

B. **High carbohydrate diet**: On high carbohydrate diet, liver is first saturated with glycogen, then any further amount of carbohydrate will be converted to triacylglycerols (lipogenesis).

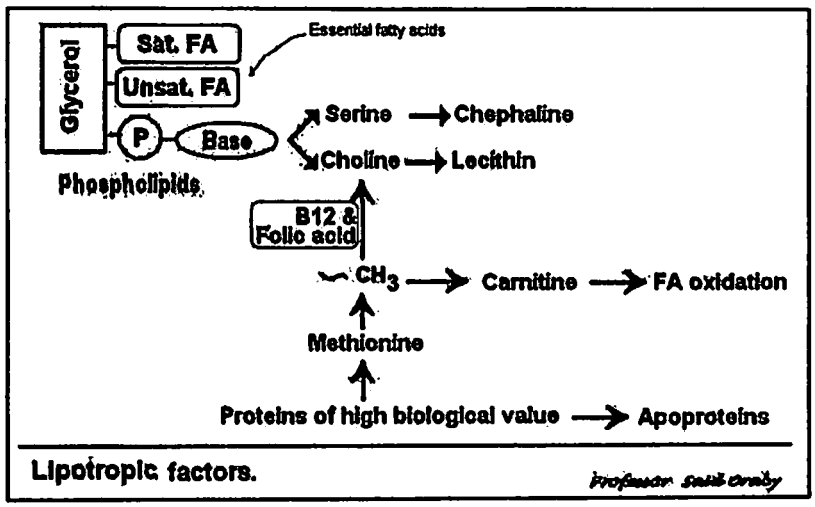
C. **Causes of under-mobilization of fat from liver to the plasma**: This is due to deficiency of any factor essential for plasma lipoproteins formation. These factors are:

1. **Deficiency of lipotropic factors:** see below
2. Decreased synthesis of apoprotein (apolipoprotein).
3. Failure in formation of phospholipids.
4. Failure in conjugation of apoproteins with triacylglycerols or phospholipids
5. Failure in secretion of lipoproteins from liver to plasma.
6. **Liver poisons:** As carbon tetrachloride, chloroform, lead and arsenic. They cause fatty liver by:
   a) Inhibition of formation of apoprotein.
   b) Inhibition of conjugation of apoprotein with lipids.
   c) Inhibition of secretion of lipoprotein.

7. **Alcoholism:** Ethanol stimulates lipogenesis, inhibiting fatty acid oxidation. This leads to accumulation of fat in liver with subsequent fatty liver.

D. Lipotropic factors:

1. **Definition:** Lipotropic factors are substances that help the mobilization of fat from the liver.
2. **They include:**
   a) **Substances enter in structure or help the formation of phospholipids.**
      1) Essential fatty acids (polyunsaturated fatty acids).
      2) Choline, enters in structure of lecithin, sphingomyelin.
      3) Inositol, enters in structure of lipositol.
      4) **Amino acids:** as
         i- Methionine which is a methyl donor essential for choline formation.
         ii- Serine which enters in the structure of cephalin.
      5) **Vitamins:** Vitamin B_{12} and folic acid have a role in synthesis and transfer of methyl group (CH_3).



Lipotropic factors.

b) **Proteins of high biological value:**
   1) These proteins include all essential amino acids which enter in structure of apoproteins in the liver.
   2) These proteins include methionine. Methionine is essential for:
      i- Choline formation.
      ii- Carnitine: a substance essential for fatty acid oxidation.

Summary of lipids pathways

	Lipogenesis	Lipolysis	FA oxidation	FA synthesis	Ketone bodies oxidation	Ketone bodies synthesis	Cholesterol
Definition	Acyl CoA + Glycerol-3-p to TG	TG to FA + glycerol	FA to Acetyl CoA to Krebs' cycle to give energy	Acetyl CoA to Palmitate	Acetacetate, β-hydroxybutyric acid and acetone		Acetyl CoA to cholesterol
Location	Cytosol Liver &adipose t.	Cytosol Of adipose tissue	Mitochondria of liver ..etc	Cytosol of liver ..etc	Mitochondria of liver	Mitochondria of extrahepatic tissue	Cytosol Liver ..
Steps	See	See	See	See	See	See	See
Functions	Synthesis of TG	----	Energy Acetyl CoA Ketone bodies	Synthesis of FA	Energy		Cell membrane Vitatmin D Bile acids Steroid Hs.
Key enzymes	Acyl CoA synthetase Glycerol-3-p DH Glycerokinase	HSTG lipase	----	Acetyl CoA carboxylase	HMG CoA synthase	Thiophrase	HMG CoA reductase
Regulation	Hormonal	Phosphorylation dephosphoryla-tion	----	Phosphorylation dephosphoryla-tion Long chain acyl CoA Hormonal	Hormonal regulation		Phosphorylation dephosphorylation Feed back inhibition Feed back regulation drugs
Diseases	-----	-----	Refsum's disease		Ketosis		Hyper and hypocholesterolemia

TG = Triacylglycerols, FA = fatty acids, DH = dehydrogenase, HSTG lipase = Hormone sensitive triacylglycerol lipase.